Core Link Technologies
OSI Layer Classification & Purpose

Document Purpose

This document outlines the key protocols and technologies used in the "Core Link Technologies" topology. It classifies each item by OSI layer and operational purpose, providing a reference for design, simulation, and troubleshooting in enterprise networks.

1. OSI Layer-Based Classification

Technology / Protocol	OSI Layer	Function
WLAN (802.11)	Layer 1 / 2	Wireless access medium
VLAN	Layer 2	Logical segmentation of broadcast domains
STP / RSTP	Layer 2	Loop prevention in switched networks
Eth-Trunk	Layer 2	Link aggregation for redundancy and throughput
LLDP Enable	Layer 2	Neighbor discovery and topology mapping
WPA / WPA2	Layer 2 / 7	Wireless encryption and authentication
Interface VLAN (SVI)	Layer 3	IP interface for VLAN routing
Inter-VLAN Routing	Layer 3	Routing between VLANs via Layer 3 device
IP Route Static	Layer 3	Manual routing configuration
RIP v1	Layer 3	Dynamic routing protocol (distance-vector)
VRRP	Layer 3	Gateway redundancy protocol
GRE Tunnel	Layer 3	Encapsulation for VPN and routing over IP
CAPWAP	Layer 3 / 7	AP control and provisioning
DHCP	Layer 7	Dynamic IP address assignment

Technology / Protocol	OSI Layer	Function
AAA	Layer 7	Authentication, Authorization, Accounting
SSL	Layer 7	Secure web encryption
SSH	Layer 7	Secure CLI access
STelnet	Layer 7	Secure Telnet (Huawei-specific)
FTP	Layer 7	File transfer protocol

2. Purpose-Based Grouping

A. Security & Access Control

- AAA – Centralized user authentication and accounting
- SSH / STelnet – Secure remote CLI access
- SSL – Encryption for web-based services
- WPA / WPA2 – Wireless encryption and user authentication

B. Routing & Redundancy

- IP Route Static – Manual route configuration
- RIP v1 – Basic dynamic routing
- VRRP – Virtual gateway failover
- GRE Tunnel – Encapsulation for secure routing over IP

C. Switching & Segmentation

- VLAN – Logical segmentation of Layer 2 domains
- Inter-VLAN Routing – Enables communication between VLANs
- Interface VLAN – Assigns IP to VLANs for routing
- STP / RSTP – Prevents Layer 2 loops
- Eth-Trunk – Aggregates multiple links
- LLDP Enable – Advertises device identity and capabilities

D. Wireless Infrastructure

- WLAN – Wireless LAN access

- CAPWAP – Centralized AP management and control
- WPA / WPA2 – Wireless security protocols

E. Services & Management

- DHCP – Automatic IP assignment
- FTP – File transfer between devices
- LLDP Enable – Device discovery and documentation

3. Simulation & Deployment Notes

- Huawei eNSP Compatibility: All protocols listed are supported in Huawei simulation environments. Use and commands for verification.
- Trunking & VLANs: Ensure interfaces are configured with and.
- VRRP: Use on both routers for redundancy.
- GRE: Tunnel interfaces require, and.
- CAPWAP: Requires AC-AP pairing and DHCP Option 43 for discovery.

4. Topology Use Case Summary

The "Core Link Technologies" topology is designed to:

- Simulate enterprise-grade Layer 2 and Layer 3 connectivity
- Demonstrate secure access and redundancy mechanisms
- Enable wireless integration and dynamic IP provisioning
- Provide a foundation for scalable, modular network design